

United International University
CSE 4833 Digital Image Processing
Class Test 1 Time: 40 Minutes Marks: 20 SET-D1

1. A satellite earth observation system deploys specialized imaging sensors along a coastal zone to monitor two distinct environmental phenomena: rapid, fine-grained wave ripple patterns along narrow beach fronts, and subtle, large-scale ocean water temperature gradient shifts (thermal pollution).
 - a. **Imaging System Alpha:** Captures images at a spatial resolution of **10,000 pixels per millimeter** but uses a **4-bit grayscale depth**.
 - b. **Imaging System Beta:** Captures images at a spatial resolution of **10 pixels per millimeter** but uses a **16-bit grayscale depth**.
 - i. Which imaging system is more likely to completely fail at resolving high-frequency spatial features, such as fine-grained wave ripples spaced **0.05 mm** apart? Provide reasoning for your answer.
[3]
 - ii. Which imaging system is more likely to fail in detecting tiny, gradual brightness variations caused by a slow **0.1degree C** ocean thermal plume shift across a wide bay? Provide reasoning for your answer.
[3]
 - iii. If bandwidth constraints limit upgrading only one parameter (spatial sampling rate or color bit depth), which would you prioritize for a real-time system detecting sharp structural edges on small sea vessels, and why? **[2]**
2. For your DIP project, you have a dataset of **handwritten Bangla characters(স্বরবর্ণ-11 + যুক্তাক্ষর-compound letters)**. There are approximately **30** most used compound letters. Each image is **64x56(hxw)** pixels, and **5-pixel blocks(only horizontal)** are used to extract features. Each block extracts **2** features. The final output should be exactly one of the characters.

How many neurons should we use in the input and output layers? What is the most suitable activation function for the output layer? Show calculations for the number of neurons in the input layer and the output layer(if any).

[4]

3. Can we use the ReLU activation function in the output layer? Justify your answer with a suitable example. [2]
4. Consider the following **5x5** grayscale image where **V = [40,150]** [6]

56	165	245	130	148
220	98	87	190	212
65	55	196	65	42
206	121	88	175	144
75	185	105	110	198

- (a) Find out and show the connectivity of all the 4-adjacent pixels.
(b) Find out and show the connectivity of all the 8-adjacent pixels.
(c) Find out and show the connectivity of all the m-adjacent pixels.